CB23F35 · LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

Hackathon Team Formation & Judging System

End Semester Examination – Individual Project

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THE PROBLEM

Hackathon judging is still done by hand

Scattered data

Registration, teams and score sheets live in separate files that are joined by hand.

No rule checks

Duplicate teams, scores outside the range and repeat judging are caught only after the event.

Late results

Totals and the leaderboard are prepared manually, so nobody sees the standing during the event.

Goal: one low-code application that enforces the rules and shows the leaderboard live.

OBJECTIVES

Five things the application had to do

1. Model participants, judges, teams and scores as linked forms
2. Automate validation, duplicate prevention, e-mail alerts and a reminder with Deluge
3. Give each role its own access, reports and a dashboard
4. Integrate Zoho Mail and the Zoho Zia Keyword Extraction AI field
5. Test with realistic data and document security, deployment and governance

WHY LOW-CODE

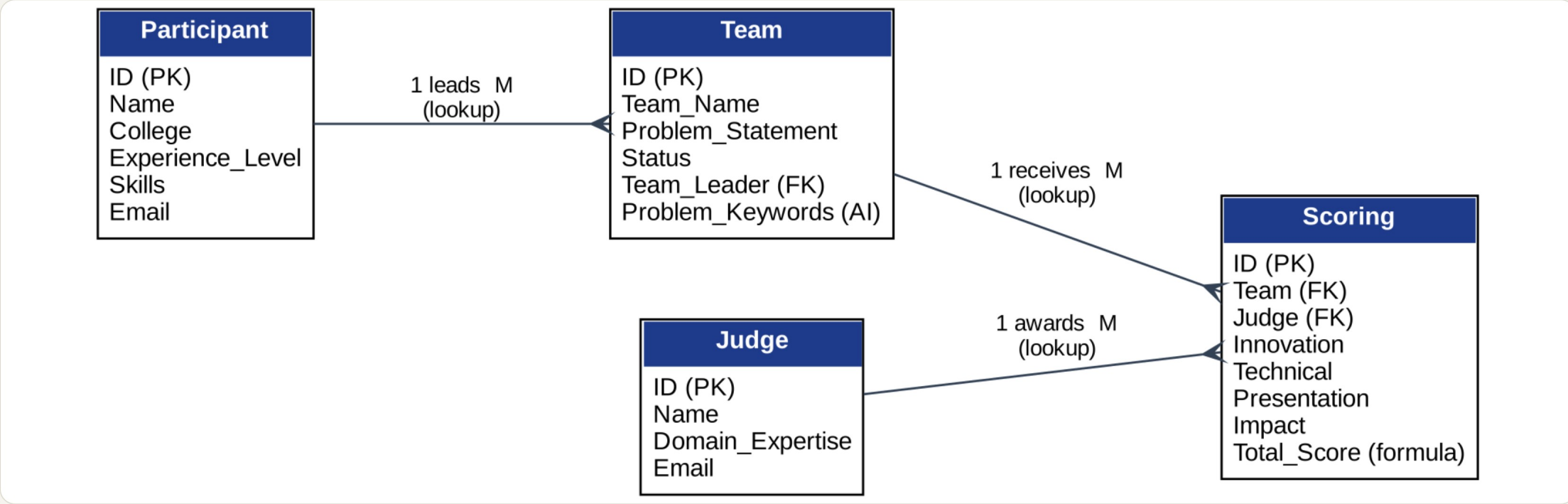
Zoho Creator instead of a custom build

Criterion	Traditional code	Zoho Creator
Development time	6–8 weeks (estimate)	About 2 weeks (estimate)
Hosting	Own server and database	Built-in cloud hosting
Access control	Coded by the developer	Roles and permission sets
Changing a rule	Edit, test, redeploy	Edit a form or script
Limits	None, at the cost of effort	Platform options only

Time figures are estimates, not measurements.

DATA MODEL

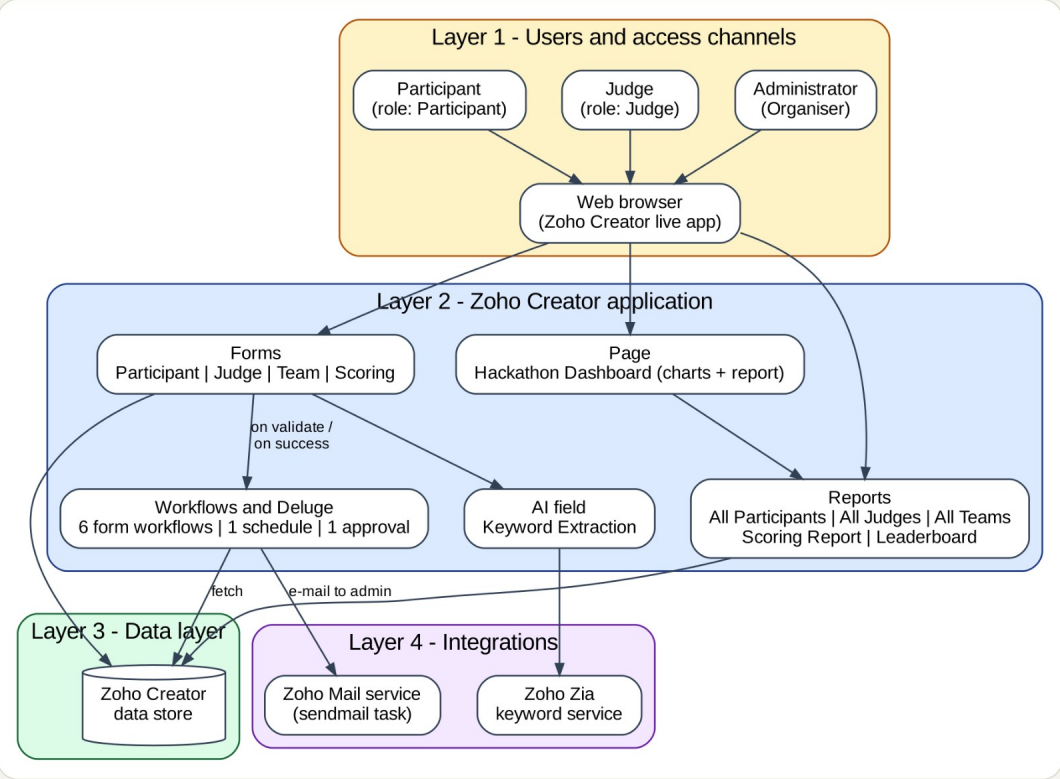
Four forms joined by lookup fields



Team.Team_Leader points to Participant. Scoring holds a Team and a Judge, so teams and judges form a many-to-many relationship.

ARCHITECTURE

Everything runs on the Zoho Creator platform



Access: Administrator, Judge and Participant use a web browser.

Application: 4 forms, 5 reports, 1 dashboard, 7 Deluge scripts, 1 approval process.

Data: the platform database, related by lookups.

Integrations: Zoho Mail and Zoho Zia. No external server or API.

FORMS

Nine field types across four forms

Participant

Name, College, Experience Level, Skills, Email (mandatory)

Judge

Name, Domain Expertise, Email (mandatory)

Team

Team Name, Problem Statement, Status, Team Leader (lookup), Problem Keywords (AI field)

Scoring

Team and Judge (lookups), four Number marks, Total Score (formula, out of 40)

Field types: Name, Single Line, Multi Line, Drop Down, Email, Number, Lookup, Formula, AI Keyword Extraction.

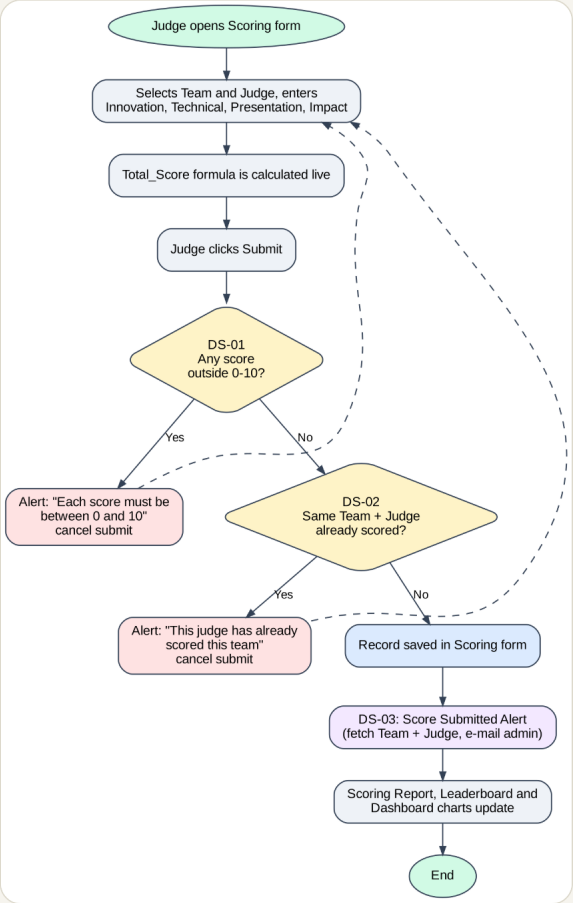
AUTOMATION

Seven Deluge scripts enforce the rules

ID	Runs on	What it does
DS-01	Scoring, validate	Rejects any mark below 0 or above 10
DS-02	Scoring, validate	Blocks a second score by the same judge for a team
DS-03	Scoring, success	E-mails the administrator the score and total
DS-04	Team, validate	Rejects a duplicate team name
DS-05	Team, edit success	E-mails the administrator when a team is Approved
DS-06	Participant, success	E-mails the administrator a new registration
DS-07	Weekly schedule	Reminder for judges to finish scoring

SCORING PROCESS

A score passes two checks before it is saved



1. The judge picks a team and enters four marks.

2. DS-01 and DS-02 validate; a bad entry shows an alert and is not saved.

3. A valid record is saved and Total Score is calculated.

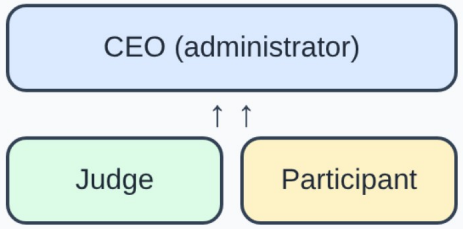
4. DS-03 e-mails the administrator.

5. Leaderboard and dashboard charts update at once.

ACCESS CONTROL

Each role sees only what it needs

Role hierarchy



Permission sets (profiles)

Permission set	Description	Component access
Judge Access	Judges view teams and enter scores	Scoring: access + view 2 reports Team: access + view All Teams Hackathon Dashboard: access
Participant Access	Participants register and view teams	Participant: access Team: access + view All Teams Hackathon Dashboard: access

Administrator: full access, receives all alerts

Judge: Scoring form, two score reports, All Teams, dashboard

Participant: Participant form, All Teams, dashboard

REPORTS AND DASHBOARD

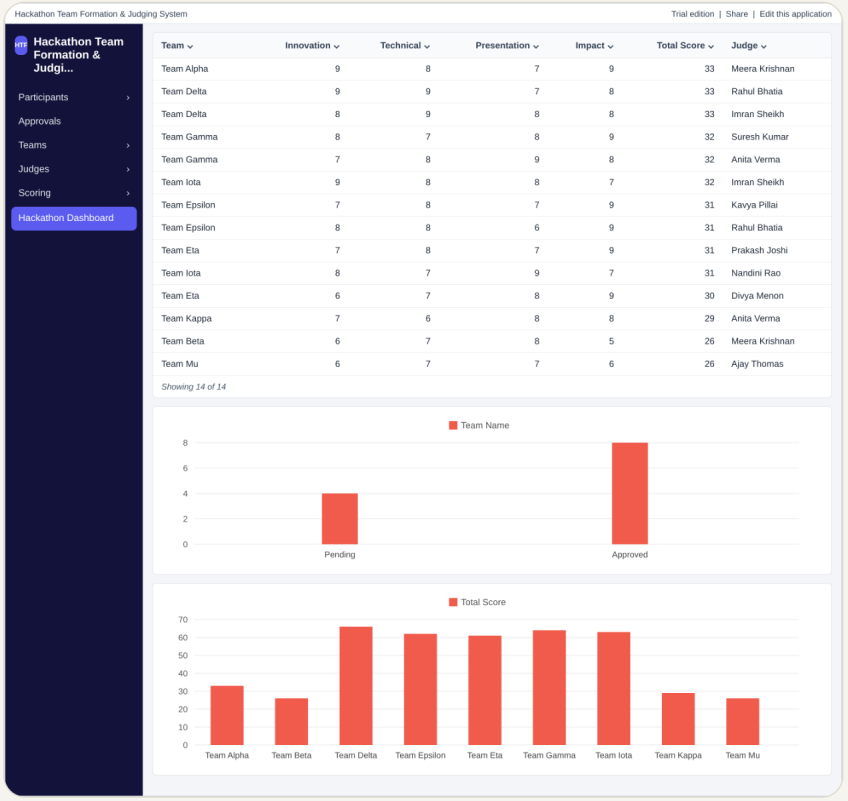
A live leaderboard and two charts

5 list reports: All Participants, All Judges, All Teams, Scoring Report, Leaderboard.

Teams by Status: 4 Pending, 8 Approved.

Total Score by Team: Team Delta leads with 66 points from two judges.

Illustration rendered from the application's data; the live application is shown in the demonstration.



AI FEATURE

Keyword Extraction tags each problem statement



Keywords came back for 8 of 12 teams. Short statements returned none and some phrases were cut, so the field is used for grouping only, never for rules.

TESTING

Tested with realistic sample data

12	10	12	14
participants	judges	teams	scores

16 test cases: 12 Pass, 4 Partial (approval run, alert delivery, weekly reminder, role login).

4 negative tests blocked: score of 11, repeat score, duplicate team, missing e-mail.

Totals: all 14 matched hand calculation. All sample e-mails use @example.com.

RESULTS

Objectives against outcomes

Objective	Status	Evidence
1. Linked data model	Met	4 forms, 3 lookups, 48 records
2. Rules and automation	Met	7 scripts; 4 negative tests blocked
3. Roles, reports, dashboard	Partly	Configured; role logins not run
4. Zoho Mail and Zia	Met	Alerts via sendmail; keywords for 8 of 12 teams
5. Testing and governance	Partly	No Stage/Production environments

LIMITATIONS

Where low-code reached its limits

Trial account

No separate Stage and Production environments; the trial expires.

AI field control

Keywords are missing for short statements and some are cut.

Notifications

E-mails go to the administrator only, since no real users were invited.

Vendor lock-in

Forms, scripts and workflows live inside Zoho Creator.

No-code covered forms, reports, charts, roles and the AI field. Low-code Deluge was needed for the rules. No pro-code was used.

CONCLUSION AND FUTURE WORK

A working event manager, ready to extend

Delivered

Four linked forms, seven scripts, an approval process, five reports and a dashboard.

Invalid scores, duplicate scores and duplicate teams are blocked, and totals and charts agree with the records.

Next steps

Send notifications to real participants and judges.

Suggest judges by matching AI keywords to expertise.

Add a team-members subform and a portal.

Move to a paid edition with Stage and Production.

Thank you

Questions and live demonstration

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